

Topics: 13.2 Derivatives and Integrals of Vector Functions; 13.3 Arc Length and Curvature; 13.4 Motion in Space: Velocity and Acceleration

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- Find the derivative of the vector function

$$\mathbf{r}(t) = \langle e^{t^2+2t}, \ln(\cos(t)), t \arctan(t) \rangle.$$

- Evaluate the definite integral

$$\int_0^1 \left(\frac{1}{t+1} \mathbf{i} + \frac{1}{t^2+1} \mathbf{j} + \frac{t}{t^2+1} \mathbf{k} \right) dt.$$

- Find a vector equation of the tangent line to the curve

$$\mathbf{r}(t) = t \cos(2t) \mathbf{i} + 4t \mathbf{j} + \sin(t) \mathbf{k}$$

at the point corresponding to $t = \frac{\pi}{4}$.

- Find the length of the curve $\mathbf{r}(t) = 2t \mathbf{i} + t^2 \mathbf{j} + \frac{1}{3}t^3 \mathbf{k}$ over the interval $0 \leq t \leq 1$.
- A particle starts at the point $(0, 0, 3)$ at time $t = 0$ and moves with position function

$$\mathbf{r}(t) = \langle 3 \sin t, 4t, 3 \cos t \rangle.$$

Find the position of the particle after it has traveled a distance of 5π units.

- Consider the vector function $\mathbf{r}(t) = \langle t, t^2, 4 \rangle$.

- Find the unit tangent vector $\mathbf{T}(t)$.
- Find the unit normal vector $\mathbf{N}(t)$.
- Find the curvature $\kappa(t)$.

- Find the vectors $\mathbf{T}$, $\mathbf{N}$, and $\mathbf{B}$ at the point $(1, \frac{2}{3}, 1)$ for the vector function

$$\mathbf{r}(t) = \langle t^2, \frac{2}{3}t^3, t \rangle.$$

- A particle moves in space with position function

$$\mathbf{r}(t) = \langle t \ln t, t, e^{-t} \rangle$$

for $t > 0$. Find the velocity, speed, and acceleration of the particle.

- The acceleration of an object is given by $\mathbf{a}(t) = t \mathbf{i} + e^t \mathbf{j} + e^{-t} \mathbf{k}$. Find the object's position function $\mathbf{r}(t)$ if the initial velocity is $\mathbf{v}(0) = \mathbf{k}$ and the initial position is $\mathbf{r}(0) = \mathbf{j} + \mathbf{k}$.
- At a point P , the velocity and acceleration vectors of a particle moving in space are $\mathbf{v} = \mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{a} = 2\mathbf{i} - 3\mathbf{k}$, respectively. Determine the curvature of the particle's path at P .

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Arc Length. If $\mathbf{r}$ is a smooth vector function on the interval $[a, b]$, then the arc length of the curve described by $\mathbf{r}$ between the points with position vectors $\mathbf{r}(a)$ and $\mathbf{r}(b)$ is

$$L = \int_a^b |\mathbf{r}'(t)| dt.$$

Unit Tangent, Unit Normal, and Binormal Vectors. Let $\mathbf{r}$ be a smooth vector function. The **unit tangent vector** at time t is defined by

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}.$$

The **(principal) unit normal vector** is defined by

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|}.$$

The **binormal vector** is defined by

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t).$$

The vectors $\mathbf{T}$, $\mathbf{N}$, and $\mathbf{B}$ are orthogonal unit vectors.

Curvature. The **curvature** of a smooth curve $\mathbf{r}(t)$ is given by

$$\kappa(t) = \left| \frac{d\mathbf{T}}{ds} \right| = \frac{|\mathbf{T}'(t)|}{|\mathbf{r}'(t)|} = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|^3}.$$

Intuitively, curvature is a scalar quantity that measures how sharply a curve bends.

Position, Velocity, and Acceleration. If a particle moves in space so that its position vector at time t is $\mathbf{r}(t)$, then the **velocity vector** of the particle is $\mathbf{v}(t) = \mathbf{r}'(t)$, its **speed** is $|\mathbf{v}(t)|$, and its **acceleration vector** is $\mathbf{a}(t) = \mathbf{r}''(t)$.

In terms of velocity and acceleration, the third curvature formula above may be written as

$$\kappa(t) = \frac{|\mathbf{v}(t) \times \mathbf{a}(t)|}{|\mathbf{v}(t)|^3}.$$

Suggested Textbook Problems

Section 13.2: 1-44, 49-52

Section 13.3: 1-30, 40-45, 51, 52

Section 13.4: 1-32, 36